An efficient path to abundant superintelligence

The last era of AI was defined by scaling, a path that is becoming unsustainable. We are ushering in a new era of AI defined by compression: inference-time intelligence that is *inexpensive*, *reliable*, and *safe*.

Scale made superintelligence possible. We are making superintelligence abundant.

Problem. The scaling paradigm has delivered incredible progress in the past half decade, taking models from simple stochastic parrots to seemingly superintelligent and capable of solving frontier problems. Notably, scaling inference-time compute has enabled increasingly complex agentic tasks, which hold the promise of delivering some of the most economically disruptive applications of AI.

However, scaling comes with its own problems: reasoning is inherently a stochastic search process that forces agents to discover solutions through compute-intensive exploration at inference time. This recurs even when the underlying structure of the solution has already been discovered, leading to issues with *cost*, *latency*, *reliability*, and *safety*.

As inference costs grow, they constrain both *further progress* and *widespread use*. Reasoning unlocked huge capabilities and opened new markets for AI. However, its inherently stochastic nature leads to the problems above, and thus reasoning is surprisingly also what limits AI from serving these markets. A new paradigm shift beyond traditional scaling is needed; Superintelligence cannot be abundant if it is prohibitively expensive, unreliable, and unsafe.

What if AI could handle consequential work 100% reliably, with logic we can inspect, at negligible cost?

Solution. When you first learn to drive, code, or prove theorems, unfamiliar steps demand slow, deliberate thought. Mistakes are common, and it is difficult to forecast the sequence of steps that you will take. With practice, these tasks become second nature: the basal ganglia turn them into reliable and energy-efficient skills. This further frees your attention up for new challenges and harder problems built on the previously practiced steps. Our scientific approach is to build this functionality for AI systems.

We are building a new architecture that, with experience, compresses compute-intensive reasoning into a new, reliable, instantaneous inference engine.

Inspired by the brain's basal ganglia's role in this transition, our system learns fast specialized inference engines from experience compressing compute-intensive reasoning. This eliminates the need for reasoning in related tasks and gives reasoning models a reliable foundation for deeper reasoning on more ambitious tasks. As early evidence, across phone-use agents "AppWorld" tasks, our specialized engine, running on ordinary CPUs, **matched in performance** an Astra-powered agent, both at approximately **~95% task success**. Average execution time fell from 26 to 0.5 seconds (**about 50× faster**), while estimated execution cost fell from $0.20 to less than $0.00001 per task (**26,000× cheaper**).

Our learned inference engines are cheap, fast, reliable and inspectable reasoning trace by design.

We are unlocking a different economics of scaling. Conventional scaling uses capital to unlock capabilities through larger models and more inference-time computation. We envision a new paradigm: *more capital* reduces the marginal cost of delivering, at fixed negligible costs, *greater intelligence* in a deterministic and interpretable manner.

Our engines are designed for local execution on general purpose hardware (CPU etc.) that gives users sovereignty over their AI deployments and data; inspectable logic provides a concrete basis for reviewing how tasks are carried out. Together, these capabilities offer a path to cheap, reliable, and interpretable superintelligence at scale—without scaling resources proportionally.

We would like to build this with you

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